Example 1.14

Evaluate : (i) $\frac{-5}{8}\times\frac{7}{3}$ (ii) $\frac{-6}{-11}\times(-4)$

Solution:

(i) $\frac{-5}{8}\times\frac{7}{3}=\frac{-5\times7}{8\times3}=\frac{-35}{24}$ (ii) $\frac{-6}{-11}\times(-4)=\frac{6}{11}\times\frac{(-4)}{1}=\frac{6\times(-4)}{11\times1}=\frac{-24}{11}$

1.3.5 Multiplicative Inverse

If the product of two rational numbers is 1, then each of them is said to be the reciprocal or the multiplicative inverse of the other.

Think

What is the multiplicative inverse of 1 and –1?

For the rational number a, its reciprocal is $\frac{1}{a}$ and vice versa since $a\times\frac{1}{a}=\frac{1}{a}\times a=1$.

For the rational number $\frac{a}{b}$, its multiplicative inverse is $\frac{b}{a}$ and vice versa since $\frac{a}{b}\times\frac{b}{a}=\frac{b}{a}\times\frac{a}{b}=1$.

1.3.6 Division

The idea of reciprocals of fractions is extended to the division of rational numbers also. To divide a given rational number by another rational number, we have to multiply the given rational number by the reciprocal of the second rational number. That is, division is simply multiplying by the multiplicative inverse of the divisor.

Example 1.15

Divide : $\frac{7}{-8}$ by $\frac{-3}{4}$

Solution:

Here, $\frac{7}{-8}\div\frac{-3}{4}=\frac{-7}{8}\times\frac{-4}{3}=\frac{7}{6}$

Try these

Divide :

(i) $\frac{-7}{3}$ by 5 (ii) 5 by $\frac{-7}{3}$ (iii) $\frac{-7}{3}$ by $\frac{35}{6}$

Note

For any non-zero b, c, and d, we have

(i) $\left(\frac{a}{b}\right)\div c=\frac{a}{bc}$ (ii) $a\div\left(\frac{b}{c}\right)=\frac{ac}{b}$ (iii) $\left(\frac{a}{b}\right)\div\left(\frac{c}{d}\right)=\frac{ad}{bc}$

1.4 Word Problems on the basic operations

Example 1.16

The sum of two rational numbers is $\frac{4}{5}$. If one number is $\frac{2}{15}$, then find the other.

Solution:

Let the other number be x.

Given, $\frac{2}{15}+x=\frac{4}{5}$

$\Rightarrow x=\frac{4}{5}-\frac{2}{15}=\frac{12-2}{15}=\frac{10}{15}\Rightarrow x=\frac{2}{3}$

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